Same Motion, Different Scores: How the Encoding of Rotations Changes Motion Metrics

Anonymous CVPR submission

Paper ID DRAFT

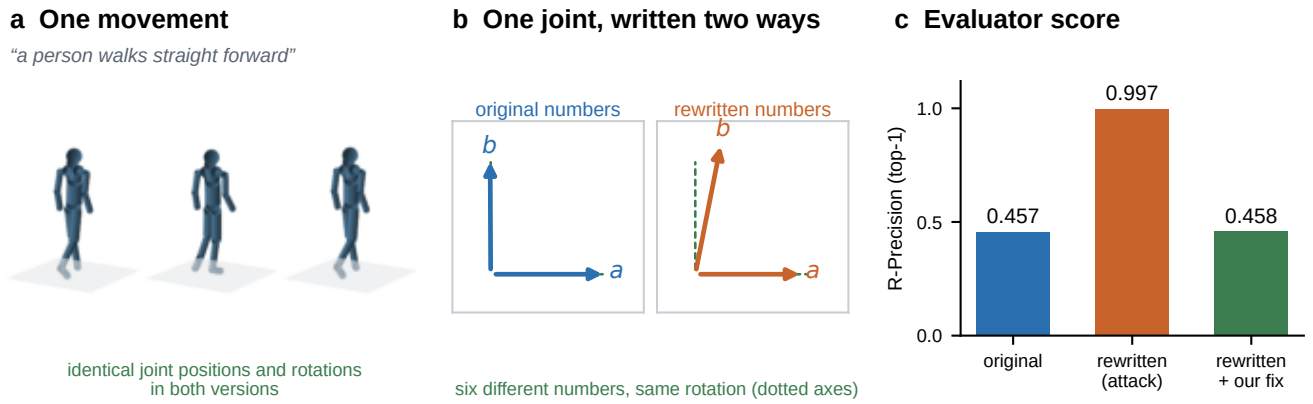


Figure 1. **Same movement, different score.** (a) A generated motion from MDM [1], rendered from its joint positions; its rewritten version has identical joint positions and rotations. (b) One joint of that motion: the six stored numbers change (vector lengths, and how far b leans toward a), but the rotation they encode, shown by the dotted axes, does not. (c) Over 1,024 generated motions, rewriting chosen with the evaluator's gradients raises the T2M evaluator's top-1 R-Precision from 0.457 (blue) to 0.997 (orange). Our fix, canonicalizing the rotations before scoring, returns the rewritten motions to 0.458 (green), the same score as the canonicalized originals.

Abstract

Generated human motion is judged by learned metrics: a neural network reads the motion and returns a score. Such a score should depend on what the body does, not on how the numbers describing it are written. We show that it does not. Joint rotations are usually stored as six numbers, of which only three matter; the other three can be changed without altering the movement at all. Changing only those three, with every joint position and rotation untouched, moves the standard text-to-motion metrics a great deal. Real motion rewritten this way gets a Fréchet distance (FID) eight times larger than a published generator's, and an attacker with access to the metric can raise caption-retrieval accuracy from 46% to over 99%. Smaller effects appear in the released outputs of ten generator settings, and the same trick more than doubles a sign-language translation score. Converting rotations to a canonical form before scoring removes the problem exactly, and we release this fix as a short drop-in function.

1. Introduction

A text-to-motion model turns a sentence such as "a person walks forward and waves" into a 3D human movement. Judging such models by eye does not scale, so the field relies on learned metrics. A neural *evaluator* turns each motion and each sentence into a vector, so that matching pairs end up close together [2, 3], and two numbers are computed from these vectors. *R-Precision* asks whether a motion's true caption ranks first among 32 candidates (higher is better). The *Fréchet distance* (FID) [4], borrowed from image generation, measures how different the vectors of generated motion are, as a group, from those of real motion (lower is better). Both are usually computed on HumanML3D [2], the standard collection of human motions paired with written descriptions. These scores rank methods and, increasingly, serve as the reward when a generator is fine-tuned [5, 6].

All of this rests on an assumption so basic that it is rarely stated: writing the same movement in two ways must give the same score.

The assumption is false. Figure 1 shows one generated motion written in two ways. Every joint position and every joint rotation is identical, and a renderer draws the same body. Yet the standard HumanML3D evaluator ranks the true caption first for 99.7% of the rewritten motions and 45.7% of the originals.

Where does this freedom come from? Joint rotations are commonly stored in the 6D representation [7]: two 3D vectors, from which the rotation is recovered by normalizing the first and making the second perpendicular to it. This throws information away on purpose. The length of each vector, and how far the second leans toward the first, never affect the result. That leaves three numbers per joint and frame that mean nothing physically. Animation tools and renderers ignore them. A neural evaluator that reads all six numbers has no reason to ignore them, and we find that it does not.

Image evaluation learned a similar lesson earlier. The same images receive different FIDs depending on the resizing filter or compression used [8], and FID can be lowered without improving the images by matching classifier statistics [9]. These findings changed how image metrics are computed and reported. Motion metrics have had no comparable audit of their input. The closest work, MARDM [10], observed that the HumanML3D evaluator leans on feature groups that generators do not need, and proposed evaluators that ignore them. Its perturbations, however, also change the movement. We ask a stricter question: can the *same* rotations, written differently, receive different scores? Answering it takes changes that alter nothing the motion represents, and a remedy that keeps the rotations, since some data, such as the sign-language features studied here, consist of nothing else. That a network reacts to inputs it never saw vary is not surprising in itself. What was unknown is how much this matters in the metrics people actually report, and whether published numbers are affected.

We study this at four levels of access to the evaluator.

- **No access.** Rewriting real motion with one fixed rule, without any caption or gradient, raises its FID to eight times that of a published generator.
- **Released outputs.** Canonicalizing the unmodified outputs of ten public generator settings changes their FID. In six, the change exceeds the margin of error their authors report.
- **Gradient access.** Choosing the rewriting with the evaluator's gradients drives retrieval to 99–100% in two different evaluators, and lowers FID by up to 59% without using any caption. An attacker who sees only the score, with no gradients, still lowers FID by up to 57%.
- **Sign language.** The same trick more than doubles the backtranslation score of generated sign-language motion, from 6.39 to 15.21 BLEU, and reproduces on real motion in two more corpora.

In every case the movement is verified to be unchanged, so a score change cannot mean better or worse motion.

The remedy is simple. Before scoring, replace each pair of vectors by its canonical form, the unit-length perpendicular pair that gives the same rotation. All equivalent writings then become the same input. Unlike discarding rotations [10], this keeps the information, needs no retraining and works for rotation-only data.

Our contributions are:

1. an audit that isolates how much learned motion evaluators depend on the way rotations are written, with controls that put the effect on a physical scale;
2. evidence that this dependence is present in public generator outputs and can be exploited without changing the motion, in two evaluators, two motion datasets and three sign-language corpora;
3. a canonical evaluation interface, with a portable implementation, that removes the dependence while keeping all rotational information.

2. Related work

Pitfalls of generative evaluation. Learned metrics are imperfect stand-ins for human judgment [11–13], and implementation details can move them as much as model changes do [8, 9]. We find a failure of the same kind in motion evaluation. Because we keep the movement fixed exactly, it does not depend on how well a metric agrees with people.

Motion representations and evaluators. The 6D rotation representation [7] and related projections [14] are standard in motion learning. The HumanML3D and KIT-ML features [2, 15] combine root motion, joint positions, velocities, 6D rotations and foot contacts, and are read by the T2M evaluator [2] and TMR [3]. Critics trained on human preferences [16] and reward fine-tuning of generators [5, 6, 17] make the consistency of such evaluators matter for training as well as reporting. MARDM [10] perturbs or removes groups of redundant features and trains evaluators on root motion and positions only. Its perturbations change represented rotations; ours do not, and our remedy keeps them.

Adversarial evaluation. Neural predictions can be redirected by universal or learned perturbations [18–20], including in sign recognizers, image-quality models, translation metrics and text-to-motion systems [21–24]. Our changes need not be small: they are invisible to the movement by construction. This allows an exact defense, whereas detectors can be evaded by adaptive attacks [25].

Sign-language production. Sign-language generators are trained on German (PHOENIX-2014T [26]), American (How2Sign [27]) and Chinese (CSL-Daily [28]) corpora. A common automatic score is *backtranslation*: a sign-to-text reader decodes the generated signing, and the text is compared with the reference sentence [29–32]. Recent work adds semantic and linguistic checks [33, 34]. We study a requirement on the reader's input, in features that contain only rotations.

3. Writing the same rotation in many ways

3.1. The freedom hidden in 6D rotations

Store one joint's rotation as two vectors $a, b \in \mathbb{R}^3$ with $a \neq 0$ and b not parallel to a. The standard conversion [7] keeps the direction of a and the part of b perpendicular to it:

$$u = a/\|a\|, \qquad b_\perp = b - (u^\top b)\, u,$$
$$v = b_\perp/\|b_\perp\|, \qquad R = [u;\, v;\, u \times v]. \tag{1}$$

Now pick any positive α, β and any real η, and rewrite the pair as

$$\mathcal{G}_{\alpha,\beta,\eta}(a, b) = (\alpha a,\ \beta b + \eta a). \tag{2}$$

The first vector still points the same way, and removing its component from the second leaves $\beta b_\perp$, which normalizes to the same v. So the rotation R is unchanged. We call this a *recoding*, and $\mathcal{C}(a, b) = (u, v)$ the *canonical form*:

$$R(\mathcal{G}(d)) = R(d), \qquad \mathcal{C}(\mathcal{G}(d)) = \mathcal{C}(d). \tag{3}$$

Conversely, every way of writing a given rotation is a recoding of its canonical form. Recodings are therefore not a few nearby alternatives: they cover all writings of the rotation, three free numbers per joint and frame.

We apply recodings only to rotation channels (Fig. 2). Every other channel and all padding is left bit-for-bit identical, and rotations, positions and forward kinematics are checked independently of any score (supplement).

3.2. Five ways to choose the recoding

A fixed rule. The same recoding for every joint and frame: $(\log\alpha, \log\beta, \eta) = (b, -b, b)$ with $b = \pm 0.1$ or ± 0.35, plus random recodings as a control. No caption, generator or gradient is involved.

Released outputs. We simply canonicalize each public generator's output and score both versions.

Targeted. With access to the evaluator's gradients, we optimize the coefficients of each frame and joint, each limited to a range set by a bound b, so that the evaluator believes the motion matches a chosen caption: its own, or another clip's.

Caption-blind. The same optimization with no caption and no test reference. The target is the statistics of the vectors of 2,048 training motions [35]. The results are saved before any reference scoring, and a second official checkpoint, which gave no gradients, scores them as well.

Learned layer. For sign language, a small network predicts the coefficients from the motion and its source sentence. It is trained once, with the generator and reader frozen, and then needs no optimization per clip [19, 20].

3.3. Closing the channel

Canonicalization removes the freedom. For any deterministic evaluator f and any recoding,

$$(f \circ \mathcal{C})(\mathcal{G}(x)) = (f \circ \mathcal{C})(x), \tag{4}$$

no matter who chose the recoding or how. Detecting unusual inputs is weaker. In the supplement, an attack that keeps both the distance to the canonical form and the feature norm unchanged still adds 4.09 BLEU to a sign-language score, while a threshold detector on that distance performs at chance.

Order matters: canonicalize the raw features first, then normalize them the way the evaluator expects. Canonicalizing numbers that were already normalized does not give the same rotation. The same interface should be used for generated and reference motion. Figure 4 shows a complete version.

4. Experimental setup

Data and evaluators. HumanML3D [2] and KIT-ML [15] are the two standard collections of 3D human motion with text descriptions. Their per-frame feature vectors (263 values for HumanML3D, 251 for KIT-ML) hold root motion, joint positions, velocities, foot contacts and the 6D rotations of 21 or 20 joints. We test two evaluators. The *T2M evaluator* [2] is the standard one for HumanML3D and reads every channel except foot contacts. Two official checkpoints of it are in use. The one released with T2M-GPT [36] reproduces the published real-motion scores (top-1 R-Precision 0.517 against the published 0.511), so we use it for the targeted attacks. The fixed rewritings, the audits of released outputs and the caption-blind and query-only attacks use the checkpoint distributed with MDM [1] (real-motion R-Precision 0.462), with the other as a replication. *TMR* [3] is a Transformer with a different architecture and its own normalization, and scores by cosine similarity. For KIT-ML we use its native evaluator.

Generators. We audit the released MDM [1], MLD [37], MotionLCM [38], T2M-GPT [36], T2M [2], MoMask [39], DiP, the diffusion planner of CLoSD [40] (which sees a

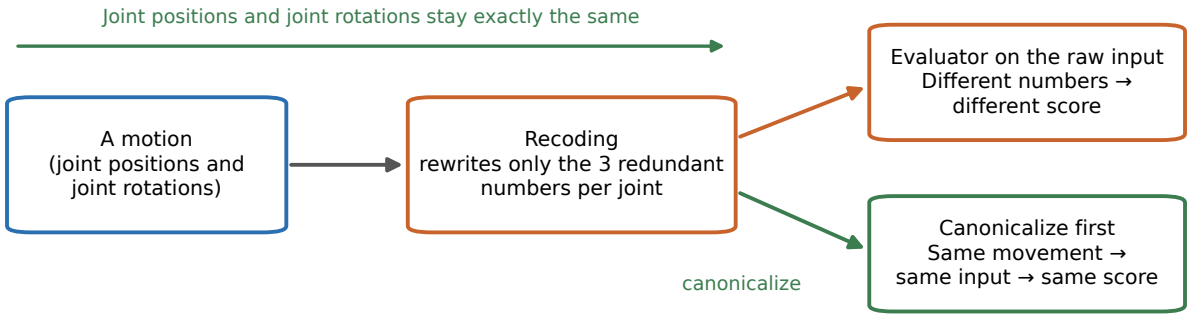


Figure 2. **What we change and what we leave alone.** A recoding rewrites only the redundant coordinates of the 6D rotations. The converter returns the same rotations, and positions, velocities, contacts and timing are untouched. A raw evaluator sees different inputs for the same movement. Canonicalizing before scoring gives every equivalent writing one input.

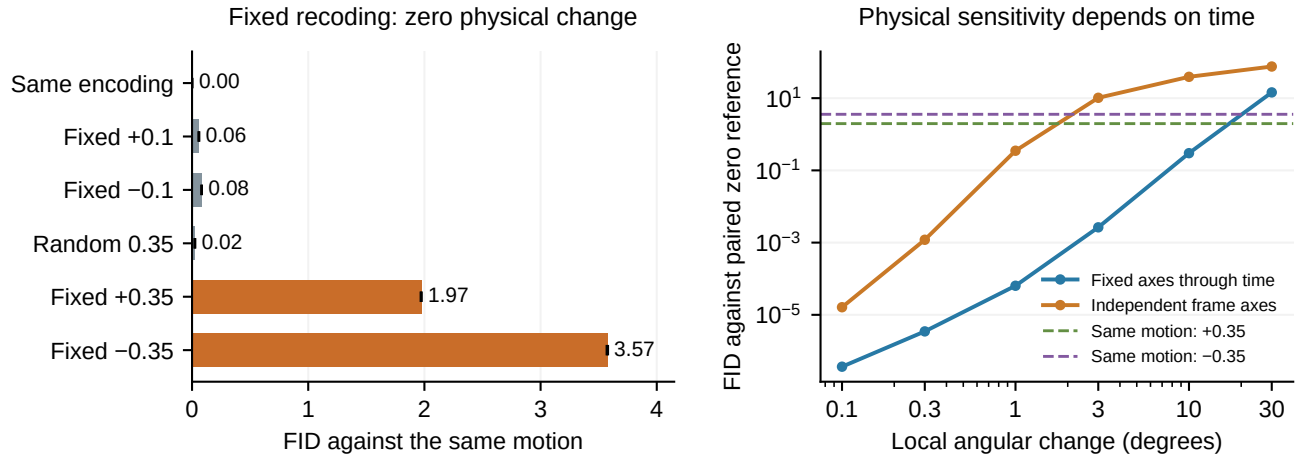


Figure 3. **Rewriting without moving the body changes FID as much as moving it.** Left: FID of real HumanML3D motion after rewriting its rotations with a fixed rule, with no captions and no optimization; bars are means over 20 draws with 95% intervals. Right: FID after really rotating the joints by a given angle, about an axis fixed through time or redrawn every frame; dashed lines mark the two largest rewritings. Everything is measured against the same reconstructed reference, so FID is zero at zero change.

```
import numpy as np
from canon6d import canonicalize
def reader_input(raw, lengths, mu, sd):
    x = canonicalize(raw, lengths)
    m = np.arange(x.shape[1])[None, :]
    m = m < lengths[:, None]
    z = (x - mu) / sd
    return z * m[..., None], m
```

Figure 4. **A canonical evaluator input.** Pass raw features and the evaluator's own normalization statistics. Use the same function for generated and reference motion.

20-frame prefix) and MotionDiffuse [41] on HumanML3D, and MoMask and MotionDiffuse on KIT-ML, using their native inference settings. Each setting uses 20 draws of the standard test protocol. Raw and canonical scores share outputs, captions and references, and intervals are paired and adjusted for multiple comparisons. These are consistency audits, not reproductions of published leaderboards.

Attacks. Targeted attacks use the same 1,024 test clips for both evaluators, 80 optimization steps and bound $b = 0.35$. The bound b limits how far each coefficient may move. Caption-blind attacks use $b = 0.1$ (and 0.35, in the supplement) on 960 MDM outputs for each of three generation seeds, and on MLD. For equal-size comparisons we rescale every change to the same distance $\kappa\sqrt{L}$ in the evaluator's normalized input space, where L is the number of frames.

Sign language. We use the released 240-value body and hand rotation features and the public SignSparK backtranslation readers [32, 42], one Sign Language Transformer [30] per corpus, on the full test sets of PHOENIX-2014T (642 clips), How2Sign (2,183) and CSL-Daily (1,162). The score is BLEU-4 [43, 44], which counts matching word

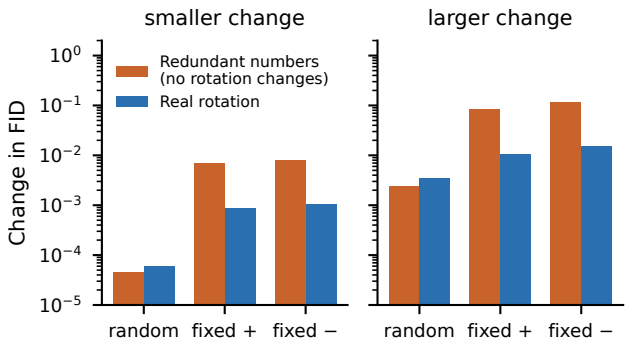


Figure 5. **Equal size, unequal response.** Change in FID of real motion for changes of equal normalized size inside the 126 rotation channels. Redundant directions (orange) leave every rotation unchanged; genuine directions (blue) rotate the joints. Means over 20 draws; intervals are narrower than the bars.

sequences between decoded text and reference (higher is better; word-level, and character-level for CSL-Daily). PHOENIX layers rewrite the output of a frozen text-to-motion model; How2Sign and CSL-Daily layers rewrite real motion. Each layer is trained with three seeds. Code, configurations and saved outputs accompany the submission.

5. Results

5.1. Rewriting alone changes the score

A fixed rule is enough. We rewrite real HumanML3D motion with one fixed rule, with no caption and no optimization. Real motion scored against itself has FID 0.001. After rewriting it scores 2.00 or 3.64, depending on the sign of the rule. The released MDM generator scores 0.455 under the same protocol, so the rewritten real motion looks four to eight times worse than a generator, although nothing about it has moved. Smaller rewritings and random ones change FID by only 0.03 to 0.08. Figure 3 (left) shows the same pattern against a paired reference (1.97 and 3.57).

The change is as large as a visible one. To put this on a physical scale we also rotate the joints for real by known angles (Fig. 3, right). The two largest rewritings score like a rotation of 1–3° applied independently to each frame, or of 10–30° about a fixed axis. A rewriting that moves nothing is scored like several degrees of real movement.

Redundant directions count more than real ones. Figure 5 compares changes of the same size within the 126 rotation channels. A fixed redundant direction moves FID by 0.08 to 0.12. A fixed direction of the same size that really rotates the joints moves it by 0.01 to 0.015, about 7.7 times less. The evaluator reacts more to numbers that carry no movement than to numbers that do. The comparison is specific to these fixed directions; random redundant changes are weaker than random real rotations.

Table 1. **Canonicalizing released outputs changes their FID.** FID of the identical motion before (Raw) and after (Canonical) canonicalization; Δ = Raw − Canonical, mean of 20 draws, and every 95% interval excludes zero. The last column gives each paper's own reported FID with its 95% margin, for scale only, since protocols differ. †: $|\Delta|$ exceeds that margin. DiP is conditioned on a short prefix.

Release	Raw	Canonical	Δ	Published
HumanML3D				
MDM [1]	0.455	0.450	$+0.005$	$0.489 \pm .047$
MLD [37]	0.445	0.449	-0.004	$0.473 \pm .013$
MotionLCM [38]	0.575	0.606	$-0.032^{\dagger}$	$0.467 \pm .012$
T2M-GPT [36]	0.183	0.149	$+0.034^{\dagger}$	$0.141 \pm .005$
T2M [2]	1.393	1.347	$+0.046^{\dagger}$	$1.087 \pm .021$
MoMask [39]	0.068	0.064	$+0.003^{\dagger}$	$0.045 \pm .002$
DiP [40]	0.367	0.365	$+0.002$	$0.228 \pm .027$
MotionDiffuse [41]	0.977	0.981	$-0.004^{\dagger}$	$0.630 \pm .001$
KIT-ML				
MoMask	0.351	0.365	$-0.014^{\dagger}$	$0.204 \pm .011$
MotionDiffuse	1.779	1.816	-0.036	$1.954 \pm .062$

A possible reason. Real HumanML3D rotations are already canonical, so the evaluator never saw the redundant numbers vary during training and was never taught to ignore them. A fixed rewriting shifts every frame the same way, which may be why it matters more than random noise. Our controls establish the difference, not this explanation.

5.2. Released generators are affected too

Generators do not output perfectly canonical rotations, so the dependence is already inside their reported scores. We canonicalize each released output, score it again, and change nothing else (Tab. 1). FID changes for all ten settings, by up to 0.046, with both signs, even among diffusion models. The published FIDs in Tab. 1 are for scale only, since protocols differ. In six of ten settings, the change is larger than the margin of error the authors report for their own FID. For scale, the published FIDs of MDM, MLD and MotionLCM are 0.489, 0.473 and 0.467, a spread of 0.022, which is smaller than the encoding-only change of MotionLCM (0.032) or T2M-GPT (0.034). An improvement of that size could therefore come from the encoding alone. With the main evaluator, the ranking of the five original generators does not change. With the second checkpoint, MotionLCM and MLD swap places in the means: MotionLCM is ahead by 0.001 FID on raw features, and MLD by 0.008 after canonicalization. Canonicalization changes the gap between them reliably (by 0.009, 95% interval [0.005, 0.013]), but under either encoding the two are statistically tied. Across all 28 generator pairs with the main checkpoint and 10 with the second, no ranking flips with intervals that exclude zero on both sides (supplement).

Table 2. **Retrieval follows the encoding, not the movement.** Top-1 R-Precision on the same 1,024 test motions before and after a caption-targeted rewriting. *True caption*: the target is the motion's own caption. *Other caption*: the target is another clip's caption and R-Precision is measured against it; "before" means no attack. Canonicalized inputs return the original scores in every row. T2M uses the checkpoint released with T2M-GPT.

		True caption		Other caption	
Evaluator	Motion	Before	After	Before	After
T2M [2]	MDM	0.457	0.997	0.028	0.916
T2M [2]	Real	0.517	0.998	0.033	0.952
TMR [3]	MDM	0.578	0.993	0.017	0.946
TMR [3]	Real	0.688	1.000	0.010	0.968

Table 3. **The backtranslation score rises while the signing does not change.** BLEU-4 on full test sets (higher is better). *Raw*: reader on the original features; *+Layer*: after the learned rewriting layer; *Canonical*: reader on canonicalized input, which is identical before and after the layer. Gain is the mean of three seeds with its paired 95% interval. CSL-Daily BLEU is character-level. Canonical need not equal Raw on real motion, because real sign features are not canonical; invariance is not the same as preserving the original score.

Corpus	Motion	Raw	+Layer	Canonical	Gain [95% CI]
PHOENIX	Generated	6.39	15.21	6.39	8.81 [7.82, 9.87]
How2Sign	Real	3.48	6.11	3.48	2.64 [2.27, 3.03]
CSL-Daily	Real	6.89	7.43	7.07	0.54 [0.14, 0.97]

Retrieval gaps are small throughout. For released models, then, the effect is a real inconsistency, comparable to small claimed improvements, but it does not overturn any ranking that was statistically supported.

5.3. An attacker can steer the evaluator

Retrieval reaches its ceiling. With the evaluator's gradients, recoding raises the true caption's top-1 rank from 0.457 to 0.997 for MDM outputs and from 0.517 to 0.998 for real motion in the T2M evaluator, and from 0.578 to 0.993 and 0.688 to 1.000 in TMR (Tab. 2). The MDM-distributed checkpoint behaves the same way (0.397 to 0.993; supplement). The evaluator is not recognizing better movement. When the target is another clip's caption, that wrong caption is retrieved 92% to 97% of the time and the true one is lost. If the same attack is run against an evaluator that canonicalizes first, nothing changes: its outputs differ by less than 10^{-5}.

No caption is needed to lower FID. Using only statistics of training motion as the target, the rewriting lowers the FID of MDM outputs by 39% to 53% across three generation seeds, and of MLD outputs by 59%. A second official checkpoint of the evaluator, which gave no gradients, still rates the same outputs 12% to 25% better (supplement). These changes are larger than the gaps between several published generators (Tab. 1), and no joint has moved.

Gradients are not needed either. We next give the attacker only one number per query: the FID of its rewritten outputs against the training motions, with no gradients, captions or test reference. A simple evolution strategy over 63 per-joint coefficients then lowers the held-out test FID by 11% to 57% after 300 queries (bound 0.1, four cohorts), and the gains carry over to the second checkpoint, which was never queried (4% to 41%). Random coefficients of the same size raise FID instead. With a larger bound the queried FID falls further, by 29% to 68%, but these gains do not carry over (supplement).

5.4. Sign language, where rotations are all there is

Sign-language features consist only of rotations, so there is no position channel to fall back on. *Backtranslation* scores a generator by letting a sign-to-text reader decode the generated signing and comparing the text with the reference sentence.

Real inputs already show the dependence. Without any attack or caption, canonicalizing released real features changes the decoded sentence for 66% of How2Sign clips (1,438 of 2,183) and 40% of CSL-Daily clips (464 of 1,162), and for none on PHOENIX. Random noise of the same size changes fewer (42% and 22%), so the readers are more sensitive to this direction than to random noise. A real physical change of the same size changes slightly more (70% and 44%), so they are sensitive to physical changes as well. These readers are also weak (3.5 BLEU on How2Sign), so this should be read as a property of these readers.

A learned layer can steer the score. A small layer reads the generated motion and its source sentence and predicts the coefficients. On PHOENIX it raises BLEU from 6.39 to 15.21, close to the 15.00 of real signing, while every canonical hypothesis is unchanged (Tab. 3). If we change only the sentence given to the layer, the decoded text moves toward that sentence: BLEU against it rises from 0.83 to 9.60. On real How2Sign and CSL-Daily motion the same construction adds 2.64 and 0.54 BLEU, with intervals that exclude zero. Optimizing each clip instead of training a layer exposes a larger channel: on PHOENIX the reader decodes an arbitrary other sentence at 85.19 BLEU (Fig. 6 and supplement).

5.5. Canonicalization closes the channel

It works exactly. Canonicalized inputs give identical evaluator outputs and identical decoded sentences for every targeted, caption-blind and learned rewriting above, as Eq. (4) predicts (numerical checks in the supplement). Evaluators that discard rotations [10] are also invariant, but they re-

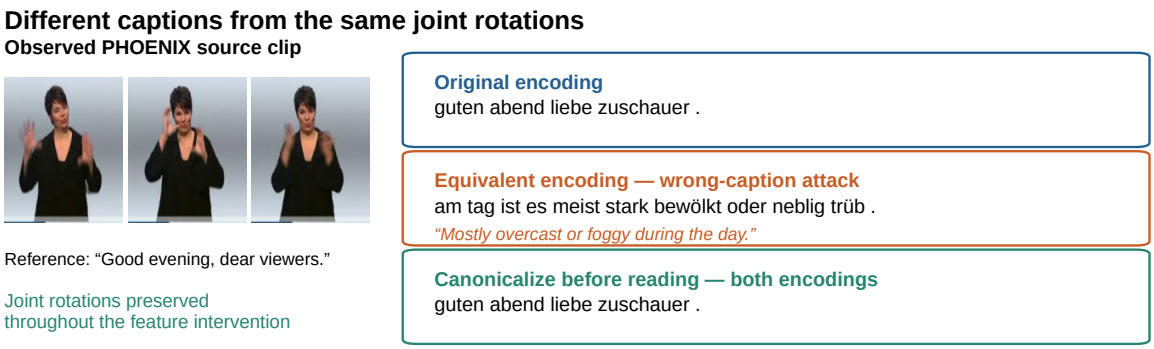


Figure 6. **A greeting becomes a weather report.** A real PHOENIX-2014T test clip (RGB frames shown for reference) and its released rotation features. Rewriting the rotations toward another sentence leaves every joint rotation unchanged, yet the reader decodes the weather report. Canonicalizing restores the original decoding for both versions. The clip was chosen for illustration; aggregate results are in Tab. 3.

place the evaluator and cannot be used on rotation-only data. Canonicalization keeps the evaluator, and its ranking of the eight HumanML3D generators agrees with that of the public essential-feature evaluators (Kendall $\tau = 0.857$).

We measure no accuracy cost. Real HumanML3D and KIT-ML rotations are already canonical, so real-motion retrieval is unchanged by construction. Real sign-language features are not. Readers fine-tuned on canonical inputs reach test BLEU within $+0.10$ (How2Sign) and -0.02 (CSL-Daily) of matched readers fine-tuned on raw inputs, with intervals that contain zero.

6. Discussion

A checklist for motion evaluation. Scores should be defined on the movement, not on its encoding. For any evaluator of rotation features we recommend: (i) name the converter and its canonical form; (ii) report each method’s raw-minus-canonical score gap, since a nonzero gap means the evaluator reads the encoding; (iii) score generated and reference motion through the same canonical interface (Fig. 4); and (iv) canonicalize learned rewards before their gradients reach a generator, because the redundant numbers are exactly what a generator can change at no cost to its motion.

Other rotation formats. The audit needs only a converter that maps many writings to one rotation. Unnormalized quaternions are indifferent to positive scale and to sign, and axis–angle vectors to adding a full turn. We have not tested these formats, but the procedure is the same: rewrite within the converter’s equivalence class, compare scores, and canonicalize before reading.

Scope. Our claims concern evaluation. We do not claim that published results were manipulated, or that standard training exploits this channel. Our query-only attacker uses a simple search over 63 coefficients; we did not try stronger black-box strategies. Canonicalization removes the dependence on rotation encoding; it does not make an evaluator accurate, and it does not address redundancy between positions and velocities or attacks that change the movement. The sign-language readers come from one released family, and their accuracy on How2Sign and CSL-Daily is low.

7. Conclusion

Learned motion evaluators give different scores to the same movement when its rotations are written differently. The effect is present in released generator outputs, can be as large as a visible physical change, and can steer retrieval, FID and backtranslation without moving a joint. Canonicalizing rotations before scoring removes it exactly and keeps all rotational information. As happened for image metrics, we hope that a precise statement of the evaluation interface becomes part of how motion results are computed and reported.

References

[1] Guy Tevet, Sigal Raab, Brian Gordon, Yoni Shafir, Daniel Cohen-Or, and Amit H. Bermano. Human motion diffusion model. In *International Conference on Learning Representations*, 2023. 1, 3, 5

[2] Chuan Guo, Shihao Zou, Xinxin Zuo, Sen Wang, Wei Ji, Xingyu Li, and Li Cheng. Generating diverse and natural 3D human motions from text. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5152–5161, 2022. 1, 2, 3, 5, 6

[3] Mathis Petrovich, Michael J. Black, and Gül Varol. TMR: Text-to-motion retrieval using contrastive 3D human motion synthesis. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 9488–9497, 2023. 1, 2, 3, 6

[4] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. GANs trained by a two time-scale update rule converge to a local nash equilibrium. In *Advances in Neural Information Processing Systems*, 2017. 1

[5] Xiaofeng Tan, Wanjiang Weng, Haodong Lei, and Hongsong Wang. EasyTune: Efficient step-aware fine-tuning for diffusion-based motion generation. In *International Conference on Learning Representations*, 2026. 1, 2

[6] Xiaofeng Tan, Wanjiang Weng, Hongsong Wang, Fang Zhao, Xin Geng, and Liang Wang. MotionRFT: Unified reinforcement fine-tuning for text-to-motion generation, 2026. 1, 2

[7] Yi Zhou, Connelly Barnes, Jingwan Lu, Jimei Yang, and Hao Li. On the continuity of rotation representations in neural networks. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5745–5753, 2019. 2, 3

[8] Gaurav Parmar, Richard Zhang, and Jun-Yan Zhu. On aliased resizing and surprising subtleties in GAN evaluation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11410–11420, 2022. 2

[9] Tuomas Kynkäänniemi, Tero Karras, Miika Aittala, Timo Aila, and Jaakko Lehtinen. The role of ImageNet classes in Fréchet inception distance. In *International Conference on Learning Representations*, 2023. 2

[10] Zichong Meng, Yiming Xie, Xiaogang Peng, Zeyu Han, and Huaizu Jiang. Rethinking diffusion for text-driven human motion generation: Redundant representations, evaluation, and masked autoregression. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 27859–27871, 2025. 2, 6

[11] Sadeep Jayasumana, Srikumar Ramalingam, Andreas Veit, Daniel Glasner, Ayan Chakrabarti, and Sanjiv Kumar. Rethinking FID: Towards a better evaluation metric for image generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9307–9315, 2024. 2

[12] George Stein, Jesse Cresswell, Rasa Hosseinzadeh, Yi Sui, Brendan Ross, Valentin Villecroze, Zhaoyan Liu, Anthony L Caterini, Eric Taylor, and Gabriel Loaiza-Ganem. Exposing flaws of generative model evaluation metrics and their unfair treatment of diffusion models. In *Advances in Neural Information Processing Systems*, pages 3732–3784. Curran Associates, Inc., 2023.

[13] Jordan Voas, Yili Wang, Qixing Huang, and Raymond Mooney. What is the best automated metric for text to motion generation? In *SIGGRAPH Asia 2023 Conference Papers*, 2023. 2

[14] Jake Levinson, Carlos Esteves, Kefan Chen, Noah Snavely, Angjoo Kanazawa, Afshin Rostamizadeh, and Ameesh Makadia. An analysis of svd for deep rotation estimation. In *Advances in Neural Information Processing Systems*, pages 22554–22565. Curran Associates, Inc., 2020. 2

[15] Matthias Plappert, Christian Mandery, and Tamim Asfour. The KIT motion-language dataset. *Big Data*, 4(4):236–252, 2016. 2, 3

[16] Haoru Wang, Wentao Zhu, Luyi Miao, Yishu Xu, Feng Gao, Qi Tian, and Yizhou Wang. Aligning human motion generation with human perceptions. In *International Conference on Learning Representations*, 2025. 2

[17] Kevin Clark, Paul Vicol, Kevin Swersky, and David Fleet. Directly fine-tuning diffusion models on differentiable rewards. In *International Conference on Learning Representations*, pages 4793–4822, 2024. 2

[18] Seyed-Mohsen Moosavi-Dezfooli, Alhussein Fawzi, Omar Fawzi, and Pascal Frossard. Universal adversarial perturbations. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017. 2

[19] Omid Poursaeed, Isay Katsman, Bicheng Gao, and Serge Belongie. Generative adversarial perturbations. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018. 3

[20] Shumeet Baluja and Ian Fischer. Learning to attack: Adversarial transformation networks. In *AAAI Conference on Artificial Intelligence*, 2018. 2, 3

[21] Yufeng Li, Meng Han, Jiahui Yu, Changting Lin, and Zhaojie Ju. Adversarial attacks on skeleton-based sign language recognition. In *International Conference on Intelligent Robotics and Applications*, pages 33–43, 2023. 2

[22] Yujia Liu, Chenxi Yang, Dingquan Li, Jianhao Ding, and Tingting Jiang. Defense against adversarial attacks on no-reference image quality models with gradient norm regularization. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 25554–25563, 2024.

[23] Yichen Huang and Timothy Baldwin. Robustness tests for automatic machine translation metrics with adversarial attacks. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 5126–5135, Singapore, 2023. Association for Computational Linguistics.

[24] Honglei Miao, Fan Ma, Ruijie Quan, Kun Zhan, and Yi Yang. Autonomous LLM-enhanced adversarial attack for text-to-motion. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 6144–6152, 2025. 2

[25] Florian Tramèr, Nicholas Carlini, Wieland Brendel, and Aleksander Madry. On adaptive attacks to adversarial example defenses. In *Advances in Neural Information Processing Systems*, 2020. 2

[26] Necati Cihan Camgoz, Simon Hadfield, Oscar Koller, Hermann Ney, and Richard Bowden. Neural sign language translation. In *IEEE Conference on Computer Vision and Pattern Recognition*, pages 7784–7793, 2018. 3

[27] Amanda Duarte, Shruti Palaskar, Lucas Ventura, Deepti Ghadiyaram, Kenneth DeHaan, Florian Metze, Jordi Torres, and Xavier Giro i Nieto. How2Sign: A large-scale multimodal dataset for continuous american sign language. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2735–2744, 2021. 3

[28] Hao Zhou, Wengang Zhou, Weizhen Qi, Junfu Pu, and Houqiang Li. Improving sign language translation with monolingual data by sign back-translation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 1316–1325, 2021. 3

[29] Ben Saunders, Necati Cihan Camgoz, and Richard Bowden. Progressive transformers for end-to-end sign language production. In *European Conference on Computer Vision*, pages 687–705, 2020. 3

[30] Necati Cihan Camgoz, Oscar Koller, Simon Hadfield, and Richard Bowden. Sign language transformers: Joint end-to-end sign language recognition and translation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10023–10033, 2020. 4

[31] Harry Walsh, Ed Fish, Ozge Mercanoglu Sincan, Mohamed Ilyes Lakhal, Richard Bowden, Neil Fox, Bencie Woll, Kepeng Wu, Zecheng Li, Weichao Zhao, Haodong Wang, Wengang Zhou, Houqiang Li, Shengeng Tang, Jiayi He, Xu Wang, Ruobei Zhang, Yaxiong Wang, Lechao Cheng, Meryem Tasyurek, Tugce Kiziltepe, and Hacer Yalim Keles. SLRTP2025 sign language production challenge: Methodology, results, and future work. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 2025.

[32] Jianhe Low, Alexandre Symeonidis-Herzig, Maksym Ivashechkin, Ozge Mercanoglu Sincan, and Richard Bowden. SignSparK: Efficient multilingual sign language production via sparse keyframe learning. In *European Conference on Computer Vision*, 2026. 3, 4

[33] Saki Imai, Mert Inan, Anthony B. Sicilia, and Malihe Alikhani. SiLVERScore: Semantically-aware embeddings for sign language generation evaluation. In *Proceedings of the 15th International Conference on Recent Advances in Natural Language Processing - Natural Language Processing in the Generative AI Era*, pages 452–461, Varna, Bulgaria, 2025. INCOMA Ltd., Shoumen, Bulgaria. 3

[34] Oliver Cory, Maksym Ivashechkin, Karahan Sahin, Oline Ranum, Jianhe Low, Edward Fish, Anton Pelykh, Ozge Mercanoglu Sincan, and Richard Bowden. BackTranslation2.0 – a linguistically motivated metric to assess sign language production. arXiv:2606.28673, 2026. Accepted at ECCV 2026. 3

[35] Rajendra Bhatia, Tanvi Jain, and Yongdo Lim. On the Bures–Wasserstein distance between positive definite matrices. *Expositiones Mathematicae*, 37(2):165–191, 2019. 3

[36] Jianrong Zhang, Yangsong Zhang, Xiaodong Cun, Yong Zhang, Hongwei Zhao, Hongtao Lu, Xi Shen, and Ying Shan. Generating human motion from textual descriptions with discrete representations. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14730–14740, 2023. 3, 5

[37] Xin Chen, Biao Jiang, Wen Liu, Zilong Huang, Bin Fu, Tao Chen, and Gang Yu. Executing your commands via motion diffusion in latent space. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 18000–18010, 2023. 3, 5

[38] Wenxun Dai, Ling-Hao Chen, Jingbo Wang, Jinpeng Liu, Bo Dai, and Yansong Tang. MotionLCM: Real-time controllable motion generation via latent consistency model. In *Computer Vision – ECCV 2024*, pages 390–408, 2025. 3, 5

[39] Chuan Guo, Yuxuan Mu, Muhammad Gohar Javed, Sen Wang, and Li Cheng. MoMask: Generative masked modeling of 3D human motions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 1900–1910, 2024. 3, 5

[40] Guy Tevet, Sigal Raab, Setareh Cohan, Daniele Reda, Zhengyi Luo, Xue Bin Peng, Amit H. Bermano, and Michiel van de Panne. CLoSD: Closing the loop between simulation and diffusion for multi-task character control. In *International Conference on Learning Representations*, 2025. 3, 5

[41] Mingyuan Zhang, Zhongang Cai, Liang Pan, Fangzhou Hong, Xinying Guo, Lei Yang, and Ziwei Liu. MotionDiffuse: Text-driven human motion generation with diffusion model. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(6):4115–4128, 2024. 4, 5

[42] LionelLow. SignSparK-BT: Public back-translation checkpoints and model card. Hugging Face, 2026. Audited release trained on re-optimized poses; model revision 0ced459f6c24d453893809eeca87e9ec80b70198 for CSL-Daily. 4

[43] Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. BLEU: A method for automatic evaluation of machine translation. In *Annual Meeting of the Association for Computational Linguistics*, pages 311–318, 2002. 4

[44] Matt Post. A call for clarity in reporting BLEU scores. In *Conference on Machine Translation: Research Papers*, pages 186–191, 2018. 4